

二、选择（10×1'）【主要考察词汇】

Unit 1

1. Computer-based artificial intelligence can function more like human intelligence when programmed to use a much faster technique for learning new objects, say two neuroscientists 【神经系统科学家】 who designed such a model that was designed to mirror human visual learning.

Neuroscience: ['njʊərəʊsaɪəns] 神经科学

2. We can get computers to learn much better from few examples by Leverage 【杠杆, 举债经营, 补充, 对...产生影响, 利用】 prior learning in a way that we think mirrors what the brain is doing.

3. It is like saying that a platypus 【鸭嘴兽】 looks a bit like a duck, a beaver 【海狸】, and a sea otter 【海獭】.

4. The anterior temporal lobe 【颞叶前部】 of the brain is thought to contain "abstract" concept representations that go beyond shape.

5. While there's been considerable work on inferring the goals and desires of agents 【具有自主性、交互性等特征的活着的计算实体】, much of this work has assumed that agents act optimally 【最理想地】 to achieve their goals.

6. This is part of our larger moonshot 【一个疯狂的想法或不大可能实现的项目】—— to reverse-engineer 【反向工程、逆向工程】 18-month-old human common sense.

7. The work builds conceptually 【在概念上】 on earlier cognitive models from Tenenbaum's group, showing how simpler inferences that children and even 10-month-old infants make about others' goals can be modeled quantitatively as a form of Bayesian inverse planning 【贝叶斯逆规划】.

Unit 2

8. TikTok's popularity over the past two years has shone the spotlight on 【成为人们关注的焦点】 ByteDance, its Beijing-based developer.

9. And ByteDance isn't done. The 60,000 people in its buzzy offices— “We are like flies,” says one former employee— crank out 【开发, 制成; 粗制滥造】 one app after another.

10. As other firms sack 【解雇】 workers amid covid-19, ByteDance is hiring 10,000 globally.

11. More haphazardly 【更随意的是, 更偶然的是】, conglomerates like Fosun and Anbang splurged on trophy assets (including ClubMed and the Waldorf Astoria hotel, respectively).

12. All this helps ensure ByteDance is not, in contrast to many Chinese tech upstarts, in hock to 【抵押给, 欠...的债】 Alibaba, Baidu and Tencent— which is handy, for Bytedance competes with China's tech titans for ad revenue.

13. And established companies, such as Daimler, a carmaker, and Goldman Sachs, a bank, are beginning to experiment with the nascent 【新兴的】 industry's products, hoping that, if they can master them, they will bestow 【给予; 使用; 安放; 安顿】 an advantage over their competitors. Much of the money is going towards building hardware.

14. By exploiting the counter-intuitive properties of quantum mechanics, they could perform some calculations (though only some) faster than any conceivable 【可想象的; 可信的】 non-quantum machine.

15. Quantum computers would also speed up the analysis of optimisation problems, which try to find the best way to juggle 【处理】 many variables in order to maximise a particular outcome.

16. The business of quantum computers 量子计算机蕴藏的商机

17. Uncertainty principals 不确定性原理

18. Quantum computing is attracting commercial interest 量子计算正在吸引商业利益

19. Big, stable quantum computers would be useful devices. 大型稳定量子计算机将拥有可观的用武之地。

Unit 3

20. This once-in-a-decade mobile phone network upgrade is expected to place us squarely in the middle of the **Internet of Things** 【(IOT) 物联网】.

21. “5G will be one of the more transformative technologies we’ve had so far,” says Karri Kuoppamaki, vice president of radio network technology and strategy at **T-Mobile** 【德国电信跨国子公司】, where he is overseeing the company’s rollout of 5G networks.

22. 5G speeds can cope with **8K image resolution** 【8K 分辨率】, which entails quadruple the amount of information of **ultra-high-definition 4K images** 【超高清 4K 图像】 and beyond even the capabilities of most TVs.

23. Other applications certain to get early 5G upgrades include mobile gaming and **virtual reality (VR)** 【虚拟现实】 apps that embed goggle-wearing users in 3D images.

24. The speed and fast response of 5G networks will also affect the kit: **VR goggles** 【VR 眼镜即 VR 头显, 虚拟现实头戴式显示设备】, for example, can be sleeker, lighter and less clunky, notes Kuoppamaki.

25. Closely related to VR is **augmented reality (AR)** 【增强现实】, where graphics and text are superimposed on real images.

26. What’s more, he adds, sensors in the machinery will constantly feed remote AI software with the monitoring data it needs to self-adjust in response to variations in product quality, and to judge when the equipment requires preventive maintenance to avoid **down-time** 【停机时间, 停工期】 and more expensive repairs.

27. **Qualcomm** 【高通公司】 wants to make 5G more accessible, more common, and faster. The company recently held its annual 5G Day event, where it made a few announcements related to the rollout of 5G, the continued development of 5G tech, and more.

28. Specifically, the company is building **Radio Access Network (RAN)** 【无线电接入网】 chips, which are designed to be used in 5G base stations for your phone (or tablet, or computer) to connect.

29. Qualcomm’s chips, of course, support the super-fast **mmWave (millimeter-wave)** 【毫米波】 networks, and the widespread **Sub-6** 【亚 6GHz 频段】 networks, so if they become widespread, you may get much better mmWave support.

30. Qualcomm also announced that in partnership with **Verizon** 【威瑞森电信】 and **Ericsson** 【爱立信公司】, it achieved a peak 5G speed of a whopping 5.06Gbps.

31. The speed was achieved with a bunch of fancy tech on a mmWave network and using what’s called **carrier aggregation** 【载波聚合】, which combines multiple channels to make for faster speeds.

32. But you should still get faster speeds than you did before, and hopefully, faster speeds than you’ll get on even the best 4G **LTE** 【(Long Term Evolution) 长期演进技术】 networks.

Unit 4

33. It may not be long before the world has to **contend with** 【与...作斗争; 和...争夺】 armies of Terminator troops.

34. We already know that China is developing biologically enhanced “super soldiers,” and now the French military has been given the **go-ahead** 【许可, 批准】 to create so-called “**augmented** 【增强的】 soldiers” by the armed forces ministry’s ethical committee.

35. France said it is seeking to enhance the cognitive, perceptive, psychological and physical capabilities of its troops through medical treatments to prevent stress and pain, substances to improve **mental resilience** 【心理韧性】 if a soldier is taken prisoner, implants and **prosthetics** 【弥补术; 修复术】.

36. Although France claims that it plans to follow humanitarian laws, armed forces minister Florence Parly **cautioned** 【警告; 劝告】 that not every country is willing to operate ethically, and it is important to be prepared for such realities.

37. Army is reportedly testing an **exoskeleton** 【外骨骼; 皮骨骼】 that can be attached to soldiers’ legs to give their productivity a 27-fold boost.

38. Red blood cells known as **respirocytes** 【呼吸细胞】 are also being explored that could prevent soldiers from getting out of breath and allow them to remain underwater for several hours.

39. Just as we’ve done with **standard cattle** 【普通牛】 to give them more back, we can do that now very precisely with humans.”

40. A report by the International Society for Military Ethics in Europe described bionic soldiers who can fight all the time, **dubbing** 【把...叫做】 them “**homo robocopus** 【人机结合体】.”

41. The report stated: “Enhanced soldiers would **be reduced** 【使变成】 to **bionic** 【仿生学的】 men, who run fast, do not need sleep, eat and drink very little, and can fight all the time.”

42. “The **hang up** 【搁置; 悬挂】 is what’s going on in the robot’s head,” she adds.

43. Perceiving **stimuli** 【刺激】 and calculating a response takes a “**boatload** 【一条船的货物; 大量】 of computation,” which limits reaction time, says Neuman, who recently graduated with a PhD from the MIT Computer Science and Artificial Intelligence Laboratory (CSAIL).

44. The method, called **robomorphic** 【机器人形态的】 computing, uses a robot’s physical layout and intended applications to generate a customized computer chip that minimizes the robot’s response time.

45. The advance could fuel a variety of robotics applications, including, potentially, frontline medical care of **contagious** 【传染的】 patients.

46. Neuman is now a postdoctoral NSF Computing Innovation **Fellow** 【研究员】 at Harvard’s School of Engineering and Applied Sciences.

47. “Current **algorithms** 【算法】 cannot be run on current CPU hardware fast enough.”

48. That means moving beyond a **standard-issue** 【常规的】 CPU processing chip that comprises a robot’s brain — with the help of hardware acceleration.

49. A commonly used hardware accelerator is the **graphics processing unit (GPU)** 【图形处理器】, a chip specialized for parallel processing.

50. These devices are **handy** 【方便的; 手边的】 for graphics because their parallel structure allows them to simultaneously process thousands of **pixels** 【像素】.

51. These matrices are “**sparse** 【稀少的; 稀疏的】,” meaning they contain many zero values that roughly correspond to movements that are impossible given a robot’s particular anatomy.

52. Similarly, your arm’s movements are limited because it can only bend at certain joints — it’s not an infinitely **pliable** 【易曲折的; 柔软的】 spaghetti noodle.

53. The resulting chip design is therefore **tailored** 【特制的; 合身的】 to maximize efficiency for the robot’s computing needs. And that customization **paid off** 【付清; 得到好处/回报】 in testing.

54. Hardware architecture designed using this method for a particular application outperformed **off-the-shelf** 【成品的; 现成的】 CPU and GPU units.

55. While Neuman’s team didn’t fabricate a specialized chip from scratch, they programmed a customizable field-programmable **gate array** 【门数组, 门阵列】 (FPGA) chip according to their system’s suggestions.

56. Despite operating at a slower **clock rate** 【时钟速率】, that chip performed eight times faster than the CPU and 86 times faster than the GPU.

57. “I was thrilled with those results,” says Neuman. “Even though we were **hamstrung** 【残废的; 做不了的】 by the lower clock speed, we made up for it by just being more efficient.”

58. “Ideally we can eventually fabricate a **custom motion-planning** 【定做的/定制的 移动规划】 chip for every robot, allowing them to quickly compute safe and efficient motions,” he says.

59. He adds that Neuman’s advance could enable robots to think faster, “**unlocking** 【开锁; 开启; 揭开】 exciting behaviors that previously would be too computationally difficult.”

60. Users will simply **drag and drop** 【拖放】 their robot’s parameters, and “out the other end comes the hardware description.

61. This research was funded by the National Science Foundation, the Computing Research Agency, the **CIFellows Project** 【计算创新研究员项目】, and the Defense Advanced Research Projects Agency.

Unit 5(未加底色的是文中标绿的词)

62. In contrast, bottlebrush polymers have additional polymers attached to the linear backbone, leading to a structure more **akin to** 【类似于】 a bottle brush you might find in your kitchen.

63. Xie's and Mukherjee's key discovery involves the self-assembly of bottlebrush polymers at the nanometer length scale, which causes a solid-to-liquid transition **in response to** 【响应, 回答, 对...作出反应】 applied pressure.

64. This material is categorized as a **yield-stress fluid** 【屈服应力流体】, meaning it begins as a semi-soft solid that **holds its shape** 【保持它的形状】, like butter or

toothpaste, but when sufficient pressure is applied, it liquefies and can be squeezed through a syringe.

65. The researchers can **tune** 【调整】 the material to flow under various amounts of pressure to match the desired processing conditions.

66. The crosslinkers can **link up** 【（与...）联合; 连接; 接合】 nearby bottlebrush polymers, **resulting in** 【导致】 a super-soft elastomer.

67. At that point, the material becomes a permanent solid -- it will no longer **liquefy** 【液化; 溶解】 under pressure -- and **exhibits extraordinary properties** 【表现出非凡的特性】.

68. Mukherjee discovered the material **by accident** 【偶然的, 不经意】, while trying to develop a material for a different project, one that would increase the amount of charge that can be stored by an **actuator** 【传动装置】.

69. "When we saw this really well-defined yield stress, it **dawned on** 【开始】 everyone collectively that we could 3D-print it," Bates said, "and that would be cool, because none of the 3D-printable materials we know of have this super-soft property."

70. "If the water **went away** 【离开; 消失; 离开（某地或某人）; 外出（尤指度假）】, then you'd just have a shapeless pile of material," said Chabinyk.

71. "With a conventional polymer, you must **figure out** 【想出; 理解; 弄清】 how to keep the right amount of water in it to **maintain its structure** 【保持其结构】, but this new material is all solid, so it will never change."

72. Mukherjee added, "We wanted the material and the printing process to be as clean and as easy as possible, so we **played a chemistry trick with** 【玩化学把戏】 solubility and self-assembly, which enabled the solvent-free process. The fact that we don't use solvent is a tremendous advantage."

73. No matter how meticulous you are in cleaning and separating your plastics, most **end up** 【（通常指意外地）最终到达; 最终; 结果; 到头来】 in the trash heap anyway.

74. Grocery bags and shrink wrap are too flimsy, **prone to** 【易于】 **getting tangled up with** 【纠缠】 other materials on a conveyor belt.

75. The difficulties of **recycling plastic into anything** 【把塑料回收成任何东西】 manufacturers want to use is a big reason why the world is **littered with** 【到处都是】 so much plastic waste, says Eric Beckman, a chemical engineer at the University of Pittsburgh.

76. Another 12 percent has been burned, and almost 80 percent has **piled up** 【堆积如山】 on land or in waterways.

77. The amount of plastic recycled in the United States has increased over the last few decades — but those levels still **pale** 【暗淡的; 微弱的】 in comparison with the amount of plastic that goes into landfills.

78. Some projects are **on the brink of** 【边缘; 濒于; 濒临】 breaking into industry; others are still just promising lab experiments.

79. "**Most plastics are like oil and water** 【大多数塑料像油和水】, " says chemist Geoffrey Coates of Cornell University.

80. Food packaging films that contain several layers of different plastic are particularly tricky to **take apart** 【拆开】.

81. The process uses a series of liquid solvents to **dissolve** 【溶; 使溶解; 解除; 终止; 解散】 individual plastic components **off** a product.

82. Dunking the remaining EVOH-PET film in a solvent called **DMSO** 【二甲基亚砜】 **stripped off** 【剥去】 the EVOH.

83. The researchers then **plucked out** 【拔出】 the remaining PET film and recovered the other two plastics from their separate solvents by mixing in "antisolvent" chemicals.

84. Those chemicals caused the plastic molecules that were **dispersed** 【(使)分散, 散开; 疏散; 驱散; 散布; 散发; 传播】 in the liquids to **bunch together into** 【聚成一团】 solid clumps that could be **fished out** 【钓鱼】.

85. When tested on **a jumble of** 【杂乱无章的】 polyethylene, PET and EVOH beads, the solvent washes recovered more than 95 percent of each material — hinting that these solvents could be used to **strip** plastic components **off** 【从...上剥离下来】 bulkier items than packaging films. 在聚乙烯, PET 和 EVOH 珠的杂物上进行测试时, 溶剂清洗回收的每种材料的回收率超过 95%, 这暗示着这些溶剂可用于将塑料组件从比包装膜更大的物品上剥离下来。

86. And existing compatibilizers are not widely used because they are not very **potent** 【有效的】 — and adding a lot of compatibilizer to a plastic blend gets expensive. 现有的增容剂由于效价不高而未被广泛使用, 并且在塑料混合物中添加大量增容剂会变得昂贵。

87. To boost viability, Coates and colleagues created a **highly potent** 【强效】 compatibilizer for polyethylene and polypropylene. 为了提高生存能力, Coates 和他的同事创造了一种用于聚乙烯和聚丙烯的高效增容剂。

88. It's as if polyethylene were made of **Legos** 【乐高积木】, and polypropylene were made of **Duplos** 【杜普洛斯】就好像聚乙烯是用乐高积木做的, 聚丙烯是用杜普洛斯做的

89. The first test of the new compatibilizer involved **welding together** 【焊接在一起】 strips of polyethylene and polypropylene.

90. **Melting down and remolding plastic is sort of like reheating pizza in the microwave — you get out basically what you put in, just not as good.** 融化和重塑塑料有点像在微波炉里加热比萨饼——你基本上得到你放进去的东西, 只是没有那么好。

91. Using heat and chemicals, it is possible to **disassemble** 【拆卸; 拆开】 polymers **into** monomers, separate those building blocks from dyes and other contaminants, and **piece** the monomers **back together into** 【把...重新拼合成...】 good-as-new plastic. 使用热量和化学物质, 可以将聚合物分解为单体, 将这些结构单元从染料和其他污染物中分离出来, 然后将单体重新拼合成新的塑料。

92. “It's not ready for **prime time** 【黄金时段】, ” he says.

93. But because it evolved to decompose plant matter, not plastic, it's not perfect. To make the enzyme better at **snipping apart** 【剪断】 PET, “we redesigned what we call the active site of the enzyme,” Marty says.

94. This involved **swapping out** 【调换】 some of the **amino acids** 【氨基酸】 along that PET docking site for others. 这涉及将 PET 对接位点上的一些氨基酸换成其他氨基酸。

95. The original enzyme had **maxed out** 【最大输出】 at about 50 percent.

96. Since this reaction takes place at a relatively mild 280° C, it happens in an orderly fashion, **snapping** long polyethylene molecules **into** 【将...捕捉到】 shorter chains that are each about 30 carbons long.

由于该反应在相对温和的 280°C 下发生, 因此有序发生, 将长的聚乙烯分子捕捉到较短的链中, 每个链长约 30 个碳。

97. Researchers in the labs of Christopher Bates, an assistant professor of materials at UC Santa Barbara, and Michael Chabynec, a professor of materials and chair of the

department, have teamed to develop the first 3D-printable “bottlebrush” **elastomer** 【[力] 弹性体, [高分子] 高弹体】.

98. That's due to the size and shape of their constituent **polymers** 【[高分子] 聚合物】 which are long, linear molecules that easily entangle like cooked spaghetti.

99. The ability to 3D-print bottlebrush elastomers makes it possible to leverage these unique mechanical properties in applications that require careful control over the dimensions of objects ranging from **biomimetic** 【仿生的】 tissue to high-sensitivity electronic devices, such as touch pads, sensors and actuators. 3D 打印牙刷弹性体的能力使得在需要仔细控制从仿生组织到高灵敏度电子设备（例如触摸板, 传感器和致动器）的对象的尺寸的应用中, 可以利用这些独特的机械性能。

100. That's important, because if the material **sags** 【中间下垂, 下凹】 during printing, the printed part will have poor structural stability.”

101. Once the object is printed, UV light is shined onto it to activate **crosslinkers** 【固化剂】 that Mukherjee synthesized and included as a part of the ink formulation.

102. The softness of a material is measured in terms of its **modulus** 【系数】, and for most elastomers, it is rather high, meaning their stiffness and elasticity are similar to those of a rubber band. 材料的柔软度是根据其模量来衡量的, 对于大多数弹性体而言, 它的硬度很高, 这意味着它们的刚度和弹性类似于橡皮筋。

103. “These super-soft elastomers might be applicable as **implants** 【移植】, ” he added.

104. The **polypropylene** 【[高分子] 聚丙烯】 in yogurt cups and other items doesn't usually get recycled either; recycling a **hodgepodge** 【混煮; 一团糟】 of polypropylene produces a dark, smelly plastic that few manufacturers will use 酸奶杯和其他物品中的聚丙烯通常也不会被回收; 回收聚丙烯大杂烩会产生深色, 有臭味的塑料, 几乎没有制造商会使用这种塑料。

105. Only two kinds of plastic are commonly recycled in the United States: the kind in plastic soda bottles, **polyethylene terephthalate** 【对苯二酸盐; 对苯二酸酯 PET】, or **PET** 【聚对苯二甲酸乙二醇酯】; and the plastic found in milk jugs and detergent containers — high-density polyethylene, or HDPE. 在美国, 通常仅回收两种塑料: 苏打瓶, 聚对苯二甲酸乙二醇酯或 PET; 以及奶罐和洗涤剂容器中的塑料-高密度聚乙烯或 HDPE。

106. With plastic collecting everywhere from the top of **Mount Everest** 【埃佛勒斯峰, 喜马拉雅山主峰之一, 中国称珠穆朗玛峰】to the bottom of the **Mariana Trench** 【马里亚纳海沟 (太平洋海底)】, there's an urgent need to reduce the amount of plastic that gets thrown away

107. Since, realistically, plastic is not going away any time soon, chemists who understand the ins and outs of all this **pesky** 【讨厌的, 麻烦的】plastic are working to make it easier to recycle and turn into higher-quality material that's useful for more things. 实际上, 由于塑料不会很快消失, 因此, 了解所有讨厌塑料的来龙去脉的化学家正在努力使其更容易回收利用, 并转变为可用于更多用途的更高质量的材料。

108. But not for items like **deodorant** 【除臭的, 防臭的】containers — where the bottle, crank and cap could all be made of different kinds of plastic.

109. The team tested the technique on a packaging film that contained polyethylene and PET, as well as a plastic oxygen barrier made of **ethylene vinyl alcohol** 【乙烯, 乙烯醇】, or EVOH, that keeps food fresh.

110. Stirring the film into a **toluene** 【甲苯】solvent first dissolved the polyethylene layer. 将薄膜搅拌到甲苯溶剂中, 首先溶解聚乙烯层。

111. Huber and colleagues next plan to look for solvents to dissolve more kinds of plastic, such as the **polystyrene** 【聚苯乙烯】in **Styrofoam** 【泡沫聚苯乙烯】.

112. Additives called **compatibilizers** 【增容剂】help different melted-down plastics blend, so that unsorted materials can be treated as one.

113. Few manufacturers can use plastic with a random **mishmash** 【混合物】of properties to make something new. 很少有制造商可以使用具有随机特性的塑料来制造新的东西。

114. The molecules that make up plastics are called polymers, which are made of smaller **monomers** 【单体】. 组成塑料的分子称为聚合物, 由较小的单体制成。

115. Microbes normally use the enzyme, called leaf-branch **compost** 【堆肥; 混合物】**cutinase** 【角质酶】, to decompose the waxy coating on plant leaves. But the cutinase is also good at breaking PET down into its monomers: **ethylene glycol** 【乙二醇】and **terephthalic acid** 【对苯二甲酸】. 微生物通常使用一种叫做叶分支堆肥角质酶的酶来分解植物叶片上的蜡状涂层。但角质酶还擅长将 PET 分解为其单体: 乙二醇和对苯二甲酸。

116. She and colleagues recently came up with a way to transform polyethylene into **alkylaromatic** 【烷基芳烃的】compounds, which can be used as biodegradable ingredients in shampoos, detergents and other products.

117. The process involves placing polyethylene inside a reaction chamber set to 280° C, with a catalyst powder containing platinum **nanoparticles** 【铂纳米颗粒】.

118. The development of a class of next-gen polymers, called PDKs, for **poly(diketoenamine)s** 【聚二酮烯胺】, was reported in Nature Chemistry in 2019.

Unit 6

119. “These results suggest that vaccines inducing durable protective immunity against SARS-CoV-2 do so by stimulating robust **germinal** 【与生殖细胞 (或胚胎) 有关的; 具生殖细胞 (或胚胎) 性质的; 根源的; 初期的; 孕育新思想的, 开创性的】center responses — a question that can be effectively answered using the **rhesus** 【猕猴; 恒河猴 (一种猕猴, 多用于实验)】model,” said Smita Iyer, assistant professor of **pathology** 【病理学】, microbiology and immunology at the UC Davis School of **Veterinary** 【兽医的】Medicine and Center for Immunology and Infectious Diseases.

病理学, 微生物学和免疫学助理教授史密塔·艾耶尔 (Smita Iyer) 表示: “这些结果表明, 疫苗可通过刺激强大的生发中心反应来诱导针对 SARS-CoV-2 的持久保护性免疫, 这一问题可以通过恒河猴模型有效地回答。” 在加州大学戴维斯分校的兽医学院和免疫学和传染病中心。

120. But an uncontrolled immune response or **“cytokine** 【细胞活素】storm” is also implicated in causing severe complications in some people. 但不受控制的免疫反应或“细胞因子风暴”也会导致一些人的严重并发症。

121. They made a type of helper cell called Th1 cells in the blood, lungs and **lymph nodes** 【淋巴结】, and produced both IgM-type antibodies and the higher-affinity IgG antibodies associated with long-term immune protection. 他们在血液, 肺部和淋巴结中制造了一种名为 Th1 细胞的辅助细胞, 并产生了与长期免疫保护相关的 IgM 型抗体和更高亲和力的 IgG 抗体。

122. Germinal centers and Tfh cells are associated with generating **plasma cells** 【浆细胞】, which remain in the body for many years to produce antibodies against **pathogens** 【病原体】the immune system has seen before. 生殖中心和 Tfh 细胞与生

成的浆细胞相关，它们在体内保留了许多年，以产生针对免疫系统以前见过的病原体的抗体。

123. Using a new gene editing strategy, researchers from Yale University can correct mutations that cause **thalassemia** 【珠蛋白生成障碍性贫血】，a form of **anemia** 【贫血症】。耶鲁大学的研究人员使用一种新的基因编辑策略，可以纠正引起地中海贫血的突变，这是一种贫血。

124. Gene editing techniques have the potential to treat blood disorders that run in families, such as thalassemia and **sickle cell anemia** 【镰状细胞血症】，but their application has been largely limited to cells in a laboratory and not living animals. 基因编辑技术具有治疗家族性血液系统疾病的潜力，例如地中海贫血和镰状细胞性贫血，但它们的应用在很大程度上仅限于实验室中的细胞，而不是活体动物。

125. To achieve gene editing in mice with thalassemia, professor of therapeutic radiology and of genetics Dr. Peter M. Glazer and his co-authors developed an alternative approach using a novel combination of **nanoparticles** 【纳米粒子】，synthetic pieces of DNA, and a simple IV injection. 为了在地中海贫血小鼠中实现基因编辑，放射线学和遗传学教授 Peter M. Glazer 博士和他的合著者开发了一种替代方法，该方法使用了纳米颗粒，DNA 合成片段和简单的 IV 注射的新型组合。

126. The cross-disciplinary research team identified a protein derived from **bone marrow** 【骨髓】 that has the ability to activate stem cells — the cells that are the most responsive to gene editing. 跨学科研究小组确定了一种源自骨髓的蛋白质，该蛋白质具有激活干细胞的能力，而干细胞是对基因编辑最敏感的细胞。

127. They combined the protein with synthetic molecules, known as PNAs, that mimic DNA and bind to the target gene to form a triple **helix** 【螺旋结构】。他们将蛋白质与合成分子（称为 PNA）结合，该分子模拟 DNA 并与靶基因结合形成三重螺旋。

128. After 140 days, they tested **hemoglobin** 【血红素；血红蛋白】 levels in the animals and found them to be normal. 140 天后，他们测试了动物的血红蛋白水平，发现它们是正常的。

129. In addition, because the research team used tiny pieces of DNA that were produced chemically, this technique avoided the type of unintended outcomes that other techniques, like CRISPR, can cause when they alter the **genome** 【基因组】。此外，由于研究团队使用的是化学合成的微小 DNA 片段，因此该技术避免了 CRISPR 等其他技术改变基因组时可能导致的意外结果。